Real Analysis: Answer to Exercise 1

Due date:

Question 1. **Solution**:

(a) Yes, $1 \in A$.
(b) No, $10 \notin A$.
(c) Yes, since $A \subseteq B$ and $B \subseteq A$.
(d) No, since $apple \in A$ but $apple \notin C$.
(e) Yes, since $C = \{1, 2\} \subseteq \{1, 2, apple\} = A$.
(f) Yes, since $C \subseteq A$ and $apple \in A \setminus C$.
(g) No, since $apple \in A$ but $apple \notin C$.
(h) No, since $pear \in D$ but $pear \notin B$.

Question 2. **Solution**:

(a) $A \cap (B \cup C) = \{1, 2, 3\} \cap \{2, 3, fish, fowl, 7, 8\} = \{2, 3\}$.
(b) $(A \cap B) \cup (A \cap C) = \{2, 3\} \cup \{2\} = \{2, 3\}$.
(c) $A \cup (B \cap C) = \{1, 2, 3\} \cup \{2\} = \{1, 2, 3\}$.
(d) $(A \cup B) \cap (A \cup C) = \{1, 2, 3, fish\} \cap \{1, 2, 3, fowl, 7, 8\} = \{1, 2, 3\}$.
(e) $C^c = U \setminus C = \{1, 3, 4, 5, 6, fish\}$.
(f) $(C^c)^c = C = \{2, fowl, 7, 8\}$.
(g) $(A \cap C)^c = \{2\}^c = \{1, 3, 4, 5, 6, 7, 8, fish, fowl\}$.
(h) $A^c \cup C^c = \{4, 5, 6, 7, 8, fish, fowl\} \cup \{1, 3, 4, 5, 6, fish\} = \{1, 3, 4, 5, 6, 7, 8, fish, fowl\}$.
(i) $(A \cup B)^c = \{1, 2, 3, fish\}^c = \{4, 5, 6, 7, 8, fowl\}$.
(j) $A^c \cap B^c = \{4, 5, 6, 7, 8, fish, fowl\} \cap \{1, 4, 5, 6, 7, 8, fowl\} = \{4, 5, 6, 7, 8, fowl\}$.

Question 3. **Solution**:

(a) $(1, 3) \cup (2, 15) = (1, 15)$.
(b) $[1, 8] \cap [4, 16] = [4, 8]$.
(c) $[66, 76] - [72, 100] = [66, 72)$.
(d) $[0, \infty) \cup (-10, 10) = (-10, \infty)$.
(e) $[27, 29] - (26, 28) = [28, 29]$.

Question 4. **Solution**: Writing $|A|$ for the number of elements of a finite set A, we have $|X \times Y| = |X| \cdot |Y| = mn$, i.e. there are mn elements in $X \times Y$.

Question 5. **Solution**: (a) There are infinitely many valid answers; here is one. Split the integers into the even and odd integers and set

$$A = \{x \in \mathbb{Z} : x/2 \in \mathbb{Z}\} \quad \text{(the even integers)},$$

$$B = \{x \in \mathbb{Z} : (x+1)/2 \in \mathbb{Z}\} \cup \{0\} \quad \text{(the odd integers, together with 0)}.$$

Then A and B each have infinitely many elements, $A \cap B = \{0\}$, and $A \cup B = \mathbb{Z}$, as required.

(b) No such sets exist. If $C \cap D = \{s, m, a, l, l\}$ then $s \in C$, hence $s \in C \cup D$; but $s \notin \{b, i, g\} = C \cup D$, a contradiction.

Question 6. **Solution**: (a) A subset of A is formed by deciding, independently for each of the m elements, whether to include it. There are 2 choices per element and m independent choices, giving 2^m subsets in total.

(b) For the first law,

$$x \in (A \cup B)^c \iff x \notin A \cup B \iff x \notin A \text{ and } x \notin B \iff x \in A^c \text{ and } x \in B^c \iff x \in A^c \cap B^c.$$

The second law is analogous: $x \in (A \cap B)^c \iff x \notin A \cap B \iff x \notin A$ or $x \notin B \iff x \in A^c$ or $x \in B^c \iff x \in A^c \cup B^c$.